

Behaving Sustainability: Cultural Cognition, Environmental Orientation, and Pro-Environmental Behavior

Introduction

Achieving environmentally sustainable outcomes requires some behavioral changes at the individual, community, and societal levels. However, not all individuals are equally likely to engage in pro-environmental behaviors, and understanding the psychological and cultural factors that drive these behaviors is important for enhancing sustainability. Environmental behaviors are widely studied (Zhu, Tong, and Song 2021), and scholars of environmental psychology often lament the so-called ‘attitude-behavior’ gap wherein environmental attitudes do not explain very much of the variance in environmental behavior (Gifford and Nilsson 2014; Kollmuss and Agyeman 2002; Stern 2000). If people care strongly about the environment, why don’t they act in a more sustainable way? The attitude-behavior gap is not because people are hypocrites, but because researchers do not yet have a complete attitude-behavior model (Kaiser, Kibbe, and Hentschke 2021).

To understand how attitudes influence behaviors, scholars use approaches such as the Value-Belief-Norm (VBN) framework, where higher-order values shape second-order beliefs that, in turn, influence behaviors []. Higher-order values form a cognitive foundation that influences how information about the world is processed and understood across multiple issues []. When considering environmental behaviors, the second-order values involve environmental cognition, such as views about the relationship of humans to the natural world and how information about the environment is understood []. In addition to environmental cognition, there are also affective feelings that individuals hold about the environment []. Together environmental cognition and affective feelings form an environmental orientation that shapes environmental behaviors []. However, to develop a complete attitude-behavior model, it is important to understand all of the various links in the chain, including the higher-order values and cognition that inform environmental orientations. Of note, the cultural worldviews measured by cultural cognition have been shown to shape environmental orientations (Nowlin 2019).

Cultural cognition was developed by Kahan and others (Kahan et al. 2007) as a tool to measure how cultural values shape beliefs about risk. It is based on cultural theory and its conceptualization of worldviews []. Cultural theory addresses how cultural values influence how people perceive and respond to risk. By measuring the cultural values of an individual, we may be able to predict how they will perceive and respond to risks

associated with certain activities or products. We are interested in how cultural cognition shapes views toward environmental issues and how those views shape behavior.

The cultural worldviews associated with cultural cognition are measured in two dimensions, a grid dimension (egalitarianism-hierarchy) and a group dimension (communitarianism-individualism). In general, those that are more egalitarian and communitarian are more concerned about environmental risks than those that are more hierarchical and individualist. Therefore, it is expected that individuals high in egalitarianism and communitarianism should engage in pro-environmental behaviors to mitigate risks to the natural environment. However, previous research shows only a weak correlation between egalitarianism and communitarianism and pro-environmental behavior (Sparks et al. 2021). Therefore, cultural cognition shapes how individuals think and process information about issues, but may not directly influence behaviors.

In this paper, we develop a theoretical framework that accounts for both the cognitive and affective influences on pro-environmental behavior. Specifically, we posit that higher-order beliefs, drawn from cultural cognition (Kahan 2012; Kahan et al. 2012; Kahan and Braman 2006), inform environmental orientations (Henry and Dietz 2012), which in turn influence behavior, both public and private. Drawing from Henry and Dietz (2012), we use the New Ecological Paradigm (NEP) as our measure of environmental cognition, and the Connectedness to Nature Scale (CNS) as the affective element of environmental orientation. Previous work failed to find that cultural cognition influence environmental behavior; however, it has been found to influence environmental cognition using the NEP (Nowlin 2019). Therefore, we hypothesize that environmental orientations mediate the relationship between cultural cognition and behavior.

Overall we find ...

Next, we develop our approach based on previous literature.

Cognition, Environmental Orientations, and Pro-Environmental Behavior

The process through values and beliefs impact behavior is often understood as a process where higher-order beliefs influence second-order beliefs, which in turn influence behavior. One of the leading frameworks that adopts this approach is the Value-Belief-Norm (VBN) framework (Stern et al. 1999). In brief, the VBN posits that underlining values inform broad environmental orientations that activate norms that in turn support pro-environmental behavior. VBN involves a sequential chain that goes from values to beliefs to behavior. Therefore, values inform attitudes, which in turn inform behaviors.

A similar set of assumptions inform a sophisticated understanding of belief systems developed by policy scholar using the Advocacy Coalition Framework (ACF) (Henry and Dietz 2012; Nohrstedt et al. 2023; Sabatier and Jenkins-Smith 1993). With this approach, beliefs systems are made up of three different types of beliefs, including deep core, policy core, and secondary beliefs. Deep core beliefs are the broadest in scope and are fundamental beliefs that inform attitudes about multiple issues. Political ideology is an example of a deep core belief. Next, are policy core beliefs, which are beliefs about particular issues. Environmental orientations would be akin to policy core beliefs. Finally, secondary aspects are the narrowest in scope and include beliefs about how best to achieve the goals outlined in policy core beliefs.

In developing our approach, we draw on the notion of a sequential chain and a hierarchical ordering of values, beliefs, and attitudes that lead from abstract values to concrete actions. Specifically, we note that deep core beliefs inform environmental orientations that, in turn, make pro-environmental behavior more (or less) likely. In addition, we separate beliefs that shape cognition from the affective and identity elements of environmental orientations. Our measure of deep core beliefs is cultural cognition, which as the name implies, are deep rooted values that influence cognition. For environmental orientations, we draw on environmental cognition, which is “the way individuals structure their thinking about environmental issues and associated political actions” (Henry and Dietz 2012, 238). We argue that environmental orientations contain a cognitive element as well as an affective element. Deep core beliefs influence both the cognitive and affective aspects of environmental orientation. Finally, deep core beliefs and environmental orientations influence pro-environmental behaviors; however, it is not clear if deep core beliefs influence behaviors directly (Markle 2019) or indirectly (Sparks et al. 2021). However, the affective element of environmental orientations tend to have the most direct influence on behaviors [].

Deep Core Beliefs: Cultural Cognition

For our measure of deep core beliefs we use cultural cognition. Cultural cognition is derived from the cultural theory of risk developed by Mary Douglas and Aaron Wildavsky (Douglas and Wildavsky 1982; Thompson, Ellis, and Wildavsky 1990). Cultural theory (CT) has been identified as a leading way to conceptualize and measure deep core beliefs (Ripberger et al. 2014). In brief, CT posits that social structures and relationships are represented by two dimensions, including a grid dimension and a group dimension. Four competing cultural worldviews arise based on their position on the grid and group dimensions. The grid dimension represents preferred social structures regarding the roles that one plays in society and the rules by which one must abide. The group dimension represents the importance of cohesive social units and the well-being of the broader community. The original formulation of CT produces four cultural worldviews based on the relative positions on the grid and group dimensions, including hierarchy (high grid, high group), egalitarianism (low grid, high group), individualism (low grid, and low group), and fatalism (high grid, low group).

Cultural worldviews provide a foundational set of deep core beliefs that inform beliefs about specific policy issues (i.e., policy core beliefs), societal controversies, and risks, among other beliefs (Jenkins-Smith et al. 2014; Nowlin 2019; Ripberger et al. 2014; Sotirov and Winkel 2016). In the context of cultural worldviews, perceptions of risk are based on the degree to which they undermine or threaten the preferred social order within each worldview. Several studies have shown a connection between cultural worldviews and risk perceptions associated with a host of environmental issues (see Xue et al. 2014), including fracking (Boudet et al. 2014), nuclear waste (Jenkins-Smith and Smith 1994), renewable energy (Tumilson et al. 2018), geoengineering (Kahan et al. 2015), high-voltage power lines (Moyer and Song 2016, 2019), and climate change (Jones 2011; Kahan et al. 2012; Nowlin 2022; Nowlin and Rabovsky 2020; Price, Walker, and Boschetti 2014), among others.

Cultural cognition draws CT and its cultural worldviews, but makes two important distinctions. The first is its focus on social and psychological mechanisms that examine how cultural worldview’s shape interpret and process information, understand risks, and take positions on policy issues. Kahan (2012) identifies five mechanisms, including cultural identity-protective cognition, where individuals conform their beliefs to those in their “group” that share cultural values; culturally biased assimilation and cultural polarization, involves

selective attention to worldview-confirming information and balanced information tends drives people of opposing worldviews further apart rather than together; cultural availability occurs as worldview-confirming events and experts are more readily noticed, credited, and recalled; cultural credibility heuristic is when people trust sources whose apparent values match their own; and cultural-identity affirmation, has been show when framing information alongside an identity-affirming rather than identity-threatening policy solution makes people more receptive to the underlying facts. Together, these mechanisms account for the cognition part of cultural cognition.

The second distinction of cultural cognition from CT is how cultural worldviews are measured empirically. As noted, the CT worldviews are based on the placement on the grid and group dimensions, and worldviews are a result of placement on both. However, cultural cognition conceives grid and group as two orthogonal dimensions with worldviews identified as being high or low on each dimension rather than a combination of the dimensions. Those high on the grid dimension are called hierarchical as they prefer a society with well-defined roles and clear lines of authority, whereas those low on the grid dimension are termed egalitarian as they value a social order based on an equal sharing of power and resources. Communitarian is the worldview that is high on the group dimension, and it represents a preference for a collectivist society. Finally, individualism is low on the group dimension preferring increased individual autonomy and responsibility (Kahan 2012; Kahan et al. 2012; Kahan and Braman 2006).

Regarding worldviews and environmental issues, in general, communitarianism, with its concern about collective well-being and egalitarianism, with its concern about the negative impacts of concentrated power tend to see threats to the natural world as a large risk. Hierarchs believe that natural resources can be well managed by experts and individualism supports the use of natural resources in private enterprise, and therefore hierarchy and individualism tend not to be concerned about environment harm. As noted, cultural worldviews have been found to be associated with attitudes about environmental issues, yet not associated with pro-environmental behaviors (Sparks et al. 2021). However, cultural cognition is associated with pro-environmental orientation, specifically the New Ecological Paradigm (NEP) (Nowlin 2019).

New Ecological Paradigm

To understand environmental behavior, scholars often turn to mental attributes related to environmental orientations. Our first measure of environmental orientations is the New Ecological Paradigm (NEP). The NEP, developed in 1978 (Dunlap and Van Liere 1978), has been widely adopted to explain environmental behavior (e.g., Stern 2000). It's original development focused on explaining the values associated with environmentalism as an emergent environmental paradigm that challenged the dominant social paradigm. Similar to cultural cognition, the NEP provides an ecological worldview that encompass values associated with the environment, and, as a worldview it includes cognition elements that shape how the world is seen []. Additionally, it is often used in the VBN as a measure of environmental beliefs (Canlas, Karpudewan, and Khan 2022). The NEP was updated in 2000, and the new measure comprises five aspects which are “the reality of the limits to growth, anti-anthropocentrism, the fragility of earth’s balance, rejection of exemptionalism, and the possibility of an ecocrisis” (Dunlap et al. 2000, 432).

The NEP is not without skeptics. One study applying scale analysis of the NEP based on a random sample of Norwegians found that the scale is not unidimensional as is often claimed (Grendstad 1999). In another

study, Sparks, Ehret, and Brick (2022) criticize the NEP for failing to adequately identify political conservatives who also hold a pro-environment orientation. Most important for our purposes is a recent study that found the NEP to be the weakest predictor of self-reported pro-environmental behavior, accounting for 12% of the variation, when compared in the same sample to the Connectedness to Nature Scale and environmentalist identity (Sparks et al. 2021). This may be because endorsement of the NEP has decreased over time (Lou et al. 2022), or it may be that feeling close and connected to nature is more proximate to behavior than more abstract, cognitive worldviews like cultural cognition and the NEP.

Connectedness to Nature

A second common instrument measuring environmental orientation is the Connectedness to Nature Scale (CNS) (Mayer and Frantz 2004). As implied by its name, the CNS seeks to measure an individual's emotional connection to nature, rather than an environmental worldview or attitude. Mayer and Franz draw inspiration from Leopold's insistence that to address environmental problems, people must feel emotionally connected to the non-human community on which we are dependent upon. The idea is that current ecological crises are a direct result of humanity losing touch with the broader community in which we live. The hypothesis can be tested by measuring an individual's affective connection to nature and determining if someone with more emotional connection correlates to more environmentally protective behaviors. They demonstrate this in their study and show that the CNS better predicts behavior (Mayer and Frantz 2004).

Going from scale development, to specific applications, one study assessed the role of personality traits on environmental behavior (Brick and Lewis 2016). The study also provided a comparison of the mediation effect of both the NEP and the CNS on environmental behaviors. The personality traits Openness and Conscientiousness were both shown to be positive predictors of environmental behavior. In both cases, the CNS had a stronger mediation effect than NEP. In other words, the CNS better predicts pro-environmental outcomes. However, these authors also note that the CNS contains language that is more consistent with liberal environmentalism, and so may fail to adequately capture the full dimensionality of environmental support across the political spectrum (Brick and Lewis 2016).

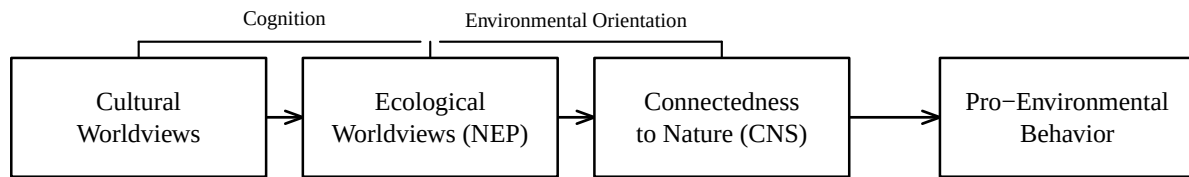
Like the NEP, the CNS is composed mostly of items that tap into politically progressive environmentalism and thus may miss conservative pro-environmental orientation (Sparks, Ehret, and Brick 2022). However, the CNS is a much stronger predictor of environmental behavior – a smaller attitude-behavior gap – than the NEP, the CNS plus controls accounting for 35% of the variance in environmental behaviors (Sparks et al. 2021).

Conceptual Model and Hypotheses

Cultural cognition can be thought of as a deep core belief, and therefore it is an antecedent to environmental orientations. Cultural-based views of risk and social-psychological tendencies likely precede the development of environment-specific cognition, attitudes, and orientations. And given the weak correlation between cultural cognition and environmental behavior it is reasonable to view this as an indirect relationship.

Taking the step from cultural worldviews to environmental orientations, we posit that environmental orientations contain both a cognitive, worldview element that can be measured by the New Ecological Paradigm (NEP) and an affective, emotional connection to nature, as measured by the Connectedness to Nature Scale (CNS). Additionally, we posit that emotional connections to nature are most predictive of pro-environmental behaviors than cultural or ecological worldviews. As noted, cultural worldviews are an antecedent to environmental orientations, and similarly, we posit that ecological worldviews are an antecedent to connectedness to nature. Finally, connectedness to nature influences both public and private pro-environmental paper. Figure 1 illustrates the conceptual model connecting cognition, based on cultural and ecological worldviews; environmental orientations, including both the ecological worldview component and the affective connection to nature component; with both public and private pro-environmental behaviors.

Figure 1: Conceptual model tested with the structural equation model: cultural cognition shapes environmental orientation—first the New Ecological Paradigm (NEP) and then the Connectedness to Nature Scale (CNS)—which in turn shapes pro-environmental behavior.



Based on the model shown in Figure 1, we posit several hypotheses regarding the relationships between cultural cognition, the NEP, the CNS, and environmental behavior. Our hypotheses include the sequence approach shown in Figure 1, where cultural worldviews influence ecological worldviews, ecological worldviews influence the connectedness to nature, and connectedness to nature influence pro-environmental behaviors. We also have direct hypotheses, where cultural and ecological worldviews are directly associated with behaviors. We expect that:

H1: Hierarchical-individualists are associated with a decrease in the ecological worldview element of environmental orientation, a decrease in the connected to nature element of environmental orientation, and fewer pro-environmental behaviors

H2: Egalitarian-communitarians are associated with an increase in the ecological worldview element of environmental orientation, an increase in the connected to nature element of environmental orientation, and more pro-environmental behaviors

H3: Environmental orientation is associated with more pro-environmental behaviors

Data and Methods

To examine the relationship between cultural cognition, environmental orientations, and behavior we draw on a sample of 501 adults living in the US purchased through Survey Sampling Inc. The survey was administered online in July 2017. The sample size was determined by cost considerations. IRB approval for the survey was

approved by ... Human Subjects Committee on April 3, 2017, protocol number 37-17-0256. Our analysis was not pre-registered.

Measures

Our dependent variables are a 12-item scale of self-reported pro-environmental behaviors, including 7 public and 5 private behaviors. Questions were taken from the 2000 Gallup Earth Day Poll. Respondents were given a list of several environmental behaviors and asked, “Which of these, if any, have you, yourself, done in the past year?” Possible responses included, “Yes, have done; No, have not done; No opinion.” Yes, responses were coded as a 1 and no/no opinion were coded as 0. For the purposes of analysis, we treat the behavior measures as observed and we added the number of self-reported behaviors together to produce a measure that ranges from 0 (no behaviors) to 12 (all of the behaviors). Table 1 shows the question wording as well as the mean and sd for each self-reported behavior.

Table 1: Self-reported pro-environmental behaviors by type (public/private), with means and standard deviations

Behavior	Type	Mean	SD
Been active in a group/organization that works to protect the environment	Public	0.19	0.39
Voted for or worked for candidates because of their environmental positions	Public	0.40	0.49
Contributed money to an environmental, conservation, or wildlife group	Public	0.33	0.47
Contacted a public official about an environmental issue	Public	0.19	0.39
Contacted a business to complain about environmentally harmful products/policies	Public	0.16	0.36
Signed a petition supporting environmental protection	Public	0.36	0.48
Attended a meeting concerning the environment	Public	0.17	0.37
Public behaviors (0-7)		1.80	1.92
Avoided using products that harm the environment	Private	0.66	0.47
Tried to use less water in your household	Private	0.75	0.43
Bought a product because it was better for the environment	Private	0.62	0.49
Voluntarily recycled newspapers, glass, aluminum, motor oil, etc.	Private	0.79	0.41
Reduced your household’s use of energy	Private	0.78	0.42
Private behaviors (0-5)		3.60	1.58
Total behaviors (0-12)		5.40	2.86

As shown in Table 1, the mean number of self-reported pro-environmental behaviors was 5.4, with a standard deviation of 2.86 in the 0 to 12 scale. For public behaviors the mean was 1.8, with a standard deviation of 1.92, and the mean number of self-reported private behaviors was 3.6, with a standard deviation of 1.58. Overall, respondents reported far more private than public behaviors. In terms of the behavior themselves, recycling had the highest mean at 0.79, which indicates that 79% of the respondents reported that they recycle, followed closely by reducing household energy use at 0.78; buying a product because it was better for the environment

had the lowest mean among private behaviors at 0.62. Among public behaviors, voting for or working for a candidate because of their environmental positions had the highest mean at 0.4, followed by signing a petition supporting environmental protection at 0.36; contacting a business to complain about environmentally harmful products/policies had the lowest mean among public behaviors at 0.16.

Our independent variables are composite measures of cultural worldviews, ecological worldviews (New Ecological Paradigm), and the Connectedness to Nature Scale. For cultural cognition, we rely on the short form of the cultural cognition survey questions which includes three questions for each of the four cultural worldviews (see Kahan 2012). Respondents were asked their agreement on a 1 (strongly disagree) to 4 (strongly agree) scale with three statements for each of cultural worldviews. To measure the New Ecological Paradigm (NEP) we used agreement with three statements on a 1 to 5 scale (1 - strongly disagree, 3 - neutral, 5 - strongly agree). Finally, for the Connectedness to Nature Scale (CNS) using respondent agreement on a 1 to 5 scale (1 - strongly disagree, 3 - neutral, 5 - strongly agree) with three statements. For both the NEP and the CNS, a higher value indicates a stronger orientation towards the environment.

Table 2 shows the question wording for each measurement item used for the latent constructs as well as the fit statistics for each of the constructs including Cronbach's α , Composite Reliability (CR), and the Average Variance Extracted (AVE). As shown in Table 2, each of the constructs show sufficient reliability (α and CR > 0.70) except for the communitarianism scale. However, CR values between 0.60 and 0.70 are acceptable for exploratory analysis (Hair et al. 2015).

Table 2: Measurement items and construct reliability (Cronbach's α , composite reliability, average variance extracted)

Construct	Statement	α	CR	AVE
Hierarchy	We have gone too far in pushing equal rights in this country.	0.83	0.83	0.62
	It seems like blacks, women, homosexuals and other groups don't want equal rights, they want special rights just for them.			
	The women's rights movement has gone too far.			
Egalitarianism	We need to dramatically reduce inequalities between the rich and poor, whites and people of color, and men and women.	0.77	0.79	0.55
	Our society would be better off if the distribution of wealth was more equal.			
	We live in a sexist society that is fundamentally set up to discriminate against women.			
Individualism	If the government spent less time trying to fix everyone's problems, we'd all be a lot better off.	0.80	0.80	0.57
	Government regulations are almost always a waste of everyone's time and money.			
	Our government tries to do too many things for too many people. We should just let people take care of themselves.			
Communitarianism	Sometimes government needs to make laws that keep people from hurting themselves.	0.61	0.64	0.38

Table 2: Measurement items and construct reliability (Cronbach’s α , composite reliability, average variance extracted)

Construct	Statement	α	CR	AVE
CNS	Government should put limits on the choices individuals can make so they don’t get in the way of what’s good for society.	0.76	0.72	0.47
	The government should do more to advance society’s goals, even if that means limiting the freedom and choices of individuals.			
	I think of the natural world as a community to which I belong.			
	I often feel a kinship with animals and plants.			
NEP	I feel as though I belong to the Earth as equally as it belongs to me.	0.70	0.73	0.48
	Humans are severely abusing the environment.			
	The so-called “ecological crisis” facing humankind has been greatly exaggerated (reverse-coded).			
	If things continue on their present course, we will soon experience a major ecological catastrophe.			

On the cultural worldview measures, we combined the scales to create dummy variables for hierarchical-individualists and egalitarian-communitarians. Respondents who scored at or above the median for both scales were coded as 1, all others were coded as 0 (similar to Kahan et al. 2012). Overall, 213 respondents (42.5%) were classified as hierarchical-individualists, and 176 respondents (35.1%) were classified as egalitarian-communitarians.

To test our hypothesis, we estimated a Structural Equation Model (SEM) using the lavaan package in R (Rosseel 2012). The dependent variable, self-reported pro-environmental behaviors, is an observed variable and the cultural cognition worldviews, the CNS, and the NEP scale are latent constructs. In addition, we examine the mediation effects of environmental orientation by examining the direct, indirect, and total effects of cultural cognition measures on pro-environmental behavior.¹

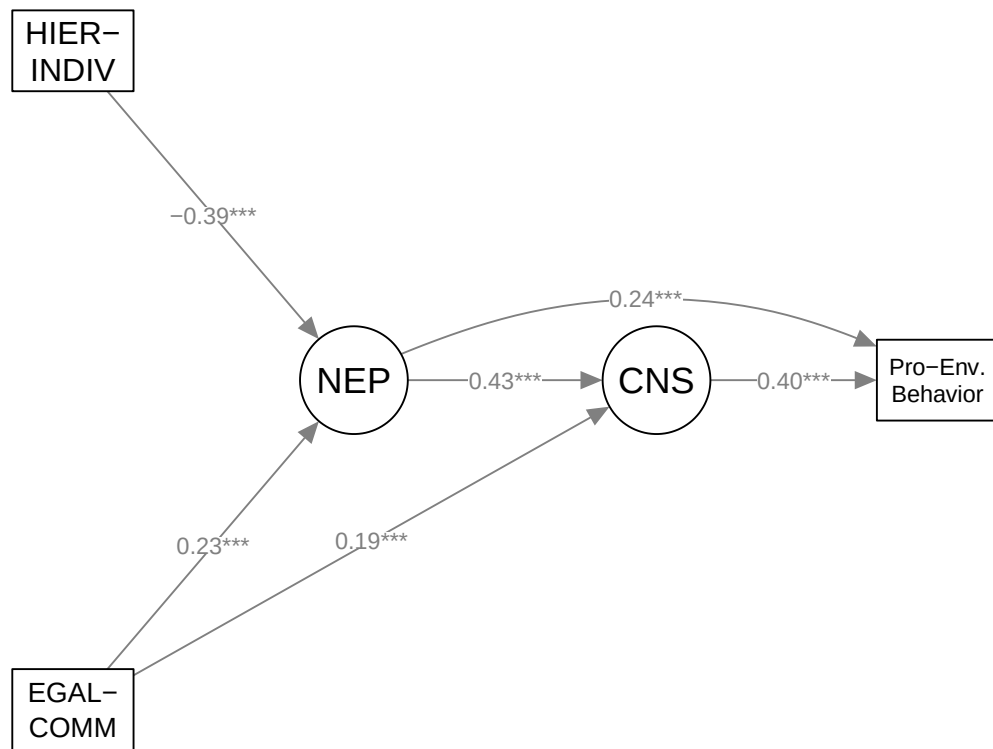
Results

The results for the SEM are shown in Figure 2. For clarity we only include the structural model and only the paths that were statistically significant at $p < 0.05$.

As can be seen in Figure 2, as expected, hierarchical-individualists were negatively associated with the NEP, whereas egalitarian-communitarians were positively associated. Also as expected, the NEP was positively associated with the CNS, and both NEP and CNS were associated with an increased number of pro-environmental

¹In the analysis we control for demographic variables, including age, gender (male=1), race (white=1), education, and income. We chose white males as the indicator variables because of the well-known white-male effect associated with risk perceptions and cultural cognition (e.g., Kahan et al. 2007). More information about the demographic variables is available in the supplemental materials.

Figure 2: Structural model of cultural worldviews, ecological worldviews, connectedness to nature, and pro-environmental behaviors.



Note: Cultural worldview groups (HIER-INDIV, EGAL-COMM) → New Ecological Paradigm → Connectedness to Nature → pro-environmental behavior (0–12 scale, public and private behaviors combined), with direct paths. Controlling for age, gender, race, education, and household income (paths not shown). The measurement (indicator) portion is omitted for clarity; standardized coefficients shown; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The full path diagram including the measurement model, and the equivalent model with the four separate worldview measures, are in the supplemental materials.

behaviors. However, there was no direct relationship to cultural worldviews and behavior. Finally, egalitarian-communitarians were more likely to feel connected to nature, therefore, they had the strongest environmental orientation as both the cognitive and affective elements were significant.

The path coefficients are standardized, which means that they equal the expected change in the standard deviation of the outcome variables for each standard deviation of the predictor, keeping all other variables constant. For example, the path coefficient from CNS to behaviors is 0.404, so holding the other variables constant, a 1-sd increase in the CNS scale is associated with a 0.404-sd increase in the number of behaviors. In terms of the number of actual predicted behaviors, as shown in Table 1, the sd of behaviors is 2.86 and 0.404 x 2.86 equals 1.155 predicted additional behaviors for each 1-sd increase in the CNS.

Standardized path coefficients allows for a comparison of strength of the various relationships. As shown, the strongest relationship is between the two environmental orientation measures, with the NEP to CNS path = 0.425. In terms of the relationship to behaviors, the CNS path is the strongest at 0.404 followed by the NEP to behavior path at 0.240. No other paths to behaviors is significant. For cultural worldviews to environmental orientations, hierarchal individualists have the strongest (negative) relationship, at -0.392, with ecological worldviews. Egalitarian communitarians have a stronger relationship to ecological worldviews at 0.231 than connectedness to nature at 0.188.

The fit statistics for the model are shown in Table 3, and include the Root Mean Square Error of Approximation (RMSEA), the Comparative Fit Index (CFI), and the Standardized Root Mean Square Residual (SRMR). Overall, the statistics indicate a reasonable fit for the model (Hu and Bentler 1999).

Table 3: Model fit statistics for the structural model

Model	Chi-sq	df	CFI	TLI	RMSEA	RMSEA 90% CI	SRMR	AIC	BIC
Full (HI/EC, + direct paths)	141.46	40	0.905	0.834	0.071	(0.059, 0.084)	0.039	10,559.8	10,715.9

Note: Model controls for age, gender, race, education, and household income. Conventional benchmarks: CFI/TLI $\geq$ 0.90 acceptable, $\geq$ 0.95 good; RMSEA $\leq$ 0.08 acceptable, $\leq$ 0.05 good; SRMR $\leq$ 0.08 good (Hu and Bentler 1999).

Next we examine the direct, indirect, and total effects of cultural worldviews on behaviors. Those results are shown in Table 4.

Table 4: Standardized decomposition of cultural worldview group effects on pro-environmental behavior.

Predictor	Direct	Serial (NEP-CNS)	Total indirect	Total
Hierarch-Individualist	0.036	-0.067***	-0.157***	-0.121**
Egalitarian-Communitarian	0.028	0.040***	0.171***	0.199***

*Note: Effects are from the full combined model, controlling for age, gender, race, education, and household income. † $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.*

For hierarchical individualists there was a total effect on behaviors of -0.121 and for egalitarian communarians it was 0.199. However, as shown in Figure 2 and Table 4, there is no direct relationship from cultural worldviews to behaviors, therefore the effects are strictly a result of indirect effects as environmental orientations appear to mediate the relationship between cultural worldviews and behaviors.

The serial effect includes the paths of the cultural worldviews to NEP, NEP to CNS, and CNS to behaviors. So, for example, for hierarchical individualist the serial effect is calculated by $-0.392 \times 0.425 \times 0.404 = -0.067$. The total indirect effect captures the paths from cultural worldviews to NEP, NEP to behaviors, cultural worldviews to CNS, and the serial effect. For hierarchical individualist that is $(-0.392 \times 0.240) + (0.010 \times 0.404) + (-0.067) = -0.094 + 0.004 - 0.067 = -0.157$. Finally, the total effect is simply the direct effect plus the total indirect effect.

Discussion

Scholars have noted a gap between attitudes and beliefs about the environment and pro-environmental behavior. Additionally, cultural worldviews derived from cultural cognition, specifically egalitarianism and communitarianism, have been shown to inform views about several environment issues as well as overall attitude toward the environment. But previous research has not found a connection between cultural worldviews and environmental behavior. In this paper, we developed and tested a conceptual model, based on previous research, where cultural worldviews influence environmental orientations and behaviors, and environmental orientations influence behaviors. Additionally, we examined the mediation effects of environmental orientations on the relationship between cultural cognition and pro-environmental behaviors. Environmental orientations were measured in terms of attitudes, using the NEP, and the emotional connection to nature, using the CNS. We used SEM to test each of our hypotheses.

We posited that egalitarianism and communitarianism would be positively associated with environmental orientations and environmental behavior (H1 a and b, H2 a and b). We found mixed results for our hypotheses. As expected, egalitarianism is associated with both the CNS and NEP; however, egalitarianism was only associated with behavior in the CNS model. However, communitarianism was associated with environmental orientation only in the CNS model and not associated with behaviors in either model.

Additionally, we posited that environmental orientations mediate the relationships between egalitarianism (H3) and communitarianism (H4) and pro-environmental behaviors. As expected, both the NEP and the CNS significantly mediate the relationship between both egalitarianism and communitarianism, and environmentally protective behavior. As far as we know, this is a novel finding. While previous studies have shown a relationship between cultural cognition and environmental orientation (Nowlin 2019) none have included a mediation analysis. Along with the results of the structural equation modeling, this helps us understand cultural cognition as an antecedent of environmental orientation and fill some of the attitude-behavior gap.

We also found that the Connectedness to Nature Scale likely had more predictive power regarding environmental behaviors than the NEP based on the model fit statistics in Table 3. This finding is in line with work showing that the CNS is a stronger predictor than the NEP (Sparks et al. 2021).

There are at least two potential reasons for the CNS displaying more power than the NEP to influence behavior, which we are only able to speculate about in the present study. First, prior work has demonstrated a slow decline in environmental concern as measured by the NEP (Lou et al. 2022). Lou and colleagues do not provide evidence for why this may be happening, only show through meta-analysis this downward trend. We are not aware of a similar study tracking CNS over time, but that may prove a fruitful avenue for further study. If it is only the NEP that is declining over time, that may indicate that the measurement instrument is becoming outdated, potentially because of shifts over time making the distinction between the dominant social paradigm and the new ecological paradigm less apt. For example, many see technological innovation as the key to unlocking a new green economy, rather than destroying the earth. Relatedly, perhaps the language used in the NEP has just become outmoded and out of line with how environmentalists currently would express environmental attitude (see Sparks, Ehret, and Brick 2022 for a discussion of this).

Second, this may be a result of the CNS and NEP measuring two distinct concepts that can be grouped under the umbrella of environmental orientation. The NEP is explicitly a measure of environmental-attitude while CNS is a measure of emotional connection to nature. For example, a recent study found that harnessing the right mix of emotions is a powerful motivator of climate action (Kovács et al. 2024).

Conclusion

In this paper, we examined the relationship between cultural cognition, environmental orientation, and pro-environmental behavior. We presented evidence that cultural cognition, egalitarianism not communitarianism, is best thought of as an indirect predictor of pro-environmental behavior. The relationship was mediated by pro-environmental orientation as measured by both the NEP and the CNS.

Because cultural cognition shapes how an individual responds to risk (e.g., Kahan et al. 2012), it was something of a puzzle as to why it only weakly correlates to behavior that protects the environment – an action that can be seen as a way to reduce environmental risk to the individual. The evidence we presented in this paper suggests that environmental orientation – whether attitude (NEP) or emotional connection (CNS) – is a missing link in this relationship.

Further, by explaining behavior with a mediation model we could close some of the so-called attitude-behavior gap. When cultural cognition and environmental orientation are aligned, people are more likely to engage in environmentally protective behavior.

Another implication of this study is further evidence that CNS is a stronger predictor of behavior than NEP. This may be useful for researchers trying to decide which measure of pro-environmental orientation to include in their questionnaire. We would like to see further testing of this finding in studies in other contexts and through meta-analytic techniques.

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